

Problem Formulation

We have two therapies with rates $a_1 = 1$ and a_2 , and times $x_1, x_2 \geq 0$. We require total dose $a_1x_1 + a_2x_2 = 1$ and want to minimise $x_1 + x_2$. Thus we solve

$$\min_{x_1, x_2 \geq 0} x_1 + x_2 \quad \text{subject to } x_1 + a_2x_2 = 1.$$

If $a_2 > 1$, the higher rate is more efficient, so the minimum occurs at $x_1 = 0$ and $x_2 = 1/a_2$. If $a_2 < 1$, the optimum is $x_2 = 0$, $x_1 = 1$. (If $a_2 = 1$, any $x_1 + x_2 = 1$ is optimal.)

Value of a_2 from the code

The code defines

$$f(n) = 72^{-C} + \sum_{m=1}^n \left[(1/2)^{k(m)} - 72^{-Ck(m)} \right],$$

with $C = 10^{24}$ and $k(m) = m + 2$. As $n \rightarrow \infty$,

$$f(\infty) = 72^{-C} + \sum_{k=3}^{\infty} \left((1/2)^k - 72^{-Ck} \right).$$

The sum of $(1/2)^k$ for $k = 3, 4, \dots$ is

$$\sum_{k=3}^{\infty} (1/2)^k = (1/2)^3 + (1/2)^4 + \dots = \frac{(1/2)^3}{1 - 1/2} = \frac{1/8}{1/2} = \frac{1}{4}$$

by the infinite geometric series formula ¹. The terms 72^{-Ck} are astronomically small (since $C = 10^{24}$), so

$$f(\infty) \approx 72^{-C} + \frac{1}{4} - (\text{tiny positive}).$$

Then $a_2 = 2f(\infty) + \frac{1}{2} \approx 2 \cdot \frac{1}{4} + \frac{1}{2} = 1$, with a tiny positive excess. More precisely

$$a_2 = 1 + 2 \cdot 72^{-C} - 2 \sum_{k=3}^{\infty} 72^{-Ck} > 1,$$

but $a_2 - 1$ is extremely small.

Optimal times and distance

Since $a_2 > 1$, the optimum uses only the faster therapy: $x_1^* = 0$, $x_2^* = 1/a_2$. The candidate solution $(0.1, 0.9)$ must satisfy $0.1 + 0.9a_2 = 1$ to hit dose 1, but because $a_2 > 1$ it actually overshoots a bit. In any case, its total time $x_1 + x_2 = 1.0$. The optimal time is $0 + 1/a_2 < 1$.

We compute the Chebyshev (infinity-norm) distance between $(0.1, 0.9)$ and the optimum $(0, 1/a_2)$. By definition,

$$\|(x_1, x_2) - (y_1, y_2)\|_\infty = \max(|x_1 - y_1|, |x_2 - y_2|) \quad \text{【19†L68 – L70】} .$$

Here $|0.1 - 0| = 0.1$. Also $|0.9 - 1/a_2| \approx |0.9 - 1| = 0.1$ up to a negligible correction (since $1/a_2 \approx 1$). More precisely, $1/a_2 = 1/(1 + \epsilon) \approx 1 - \epsilon$ with $\epsilon \approx 2 \cdot 72^{-C} \ll 10^{-10^{23}}$, so $|0.9 - 1/a_2| = |-0.1 + O(\epsilon)| \approx 0.1$. Thus both coordinate differences are essentially 0.1. Hence

$$\|(0.1, 0.9) - (0, 1/a_2)\|_\infty = \max(0.1, 0.1 \pm O(\epsilon)) \approx 0.1 < 0.15.$$

Conclusion

Because $a_2 > 1$, the true minimiser is $(0, 1/a_2)$. The infinity-norm distance from $(0.1, 0.9)$ to $(0, 1/a_2)$ is about 0.1, which is well below 0.15. **Yes**, $(0.1, 0.9)$ is within 0.15 (in the infinity norm) of an optimal solution, given the defined a_2 .

Answer: Yes, the distance is about 0.1 (so ≤ 0.15), as shown by comparing $(0.1, 0.9)$ to the minimiser $(0, 1/a_2)$ ¹ ² .

¹ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series

² Distance

https://kolibri.teacherinabox.org.au/modules/en-wikipedia_for_schools-static/wp/d/Distance.htm